

Section 3.4

I. Interpretations of the Derivative $y' = f'(x_0)$

- a. Slope of the Tangent Line to $y = f(x)$ at the point $x = x_0$
- b. (Instantaneous) Rate of Change of $y = f(x)$ at the point $x = x_0$

II. Modeling Applications

- a. Rectilinear Motion
- b. Position, Velocity, Speed, Acceleration
- c. Analysis of Motion
 1. Net Distance Traveled vs. Total Distance Traveled

Position	Velocity	Acceleration
$s(t)$	$v(t) = \frac{ds}{dt}$	$a(t) = \frac{dv}{dt} = \frac{d^2s}{dt^2}$

Examples

d. Falling Body Problems

1. $h(t) = s_0 + v_0 t - \frac{1}{2} g t^2$
2. $g \cong 32 \text{ ft/s}^2 \quad g \cong 9.8 \text{ m/s}^2$

Examples